

48V Electrical Architecture

Document role

- This file owns the final 48V electrical architecture for the active build baseline.
- Use it for house-bank structure, alternator-path direction, shutdown logic, and manual control-path intent.
- Keep full conductor schedules, fuse matrices, and layout-level implementation detail in docs/implementation/ .
- Keep unresolved vendor gates, risk state, and follow-up closure items in docs/core/TRACKING.md .
- Keep broad project sequencing and day-to-day execution framing in docs/core/PROJECT.md or the active plan docs.

As-of date: 2026-05-28

Purpose: hold the finalized, concise 48V house and alternator architecture in one place so wiring, protection, shutdown behavior, and BOM references are easy to understand without re-reading the historical trade studies.

Related docs:

- [Electrical topology diagram](#)
- [Electrical fuse schedule](#)
- [Systems](#)
- [Tracking](#)
- [Estimated BOM](#)

Final direction

- House architecture stays 48V core with 3x 51.2V 100Ah batteries in parallel.
- Alternator charging is the dedicated 48V secondary alternator path; obsolete pre-Mechman charger hardware is removed from active planning.
- Protection baseline is Mechman + WS500 + Balmar APM-48 .
- Manual alternator-charge enable/disable is through Ford Upfitter Switch #3 .
- Upfitter #3 is a low-current control signal only. It does not carry alternator output current.
- WS500 white Feature-In is reserved for future fault-interlock work, not required in Phase 1.

Current commissioning state (2026-05-27)

- 48V bus has been live-tested: owner measured 55.5V throughout the system, including at the MultiPlus.
- MultiPlus-II DC/inverter mode has been switched on with inverter light illuminated, slight normal hum, and no reported error lights.
- SmartShunt and Orion-Tr Smart are visible in VictronConnect.
- Cerbo GX access point/remote-console workflow is active; Cerbo power is a small inline fused feed from the 48V system side and MultiPlus communication is via VE.Bus RJ45.
- Shore charging has been short-tested through the MultiPlus at household-outlet current limits. This proves basic AC-in/charger function, not final charge-profile correctness.
- Do not treat the charger as commissioned for unattended/sustained use until the MultiPlus LiFePO4 charge profile is programmed/verified by live test. Current commissioning target is 56.8V absorption/charge, 54.0V float, short absorption dwell, equalization off, conservative LiFePO4 storage behavior, and source-current limit matched to the actual shore circuit. The Dumfume manual's 58.4V +/-0.2V value is documented, but because the same manual also lists 58.4V as charge-limit/over-charge protection voltage, it is not the active routine charger target while top-end BMS protection behavior is being seen. DVCC remains disabled unless a documented BMS/GX control path is added.
- AC-out branch/GFCI commissioning and secondary-alternator commissioning are still separate future gates.

Locked component set

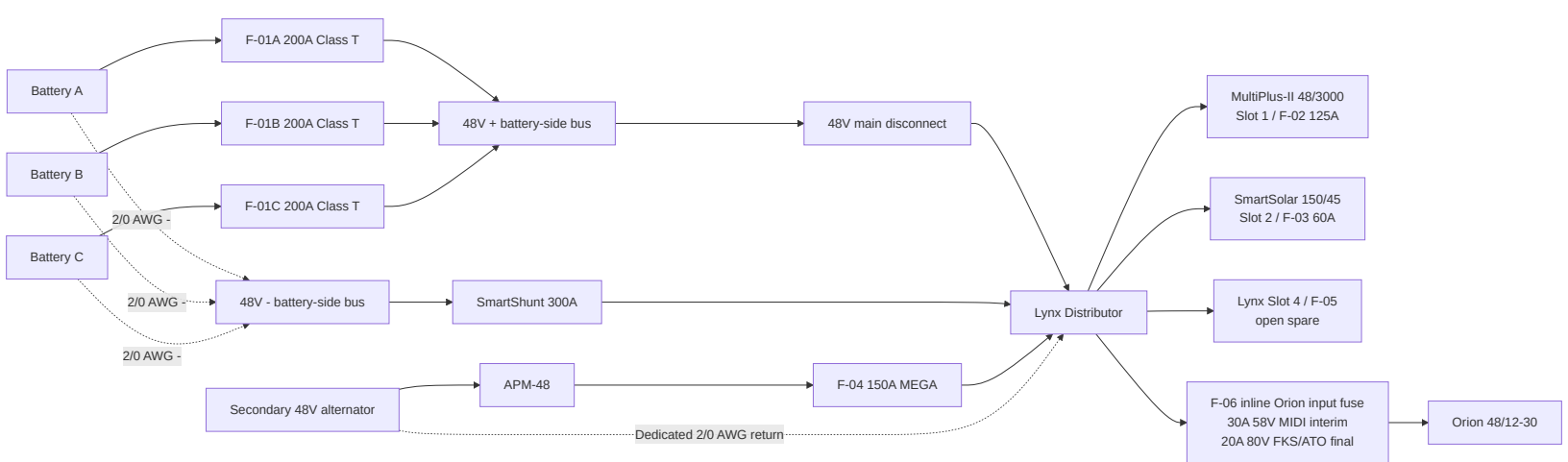
Function	Locked baseline	BOM row(s)
House batteries	3x Dumfume 51.2V 100Ah LiFePO4	3
Main house disconnect	Victron 275A battery switch	5
Main fused distribution	Victron Lynx Distributor M10	6

Battery branch protection	F-01A/B/C 200A Class T; manual-backed battery limit is 200A max continuous discharge per battery, with terminal/manufactureur fuse guidance still not separately specified; owner confirmed 3 holders and 4 slow-blow Class T fuses total	7
Inverter/charger	MultiPlus-II 48/3000/35-50	12
48V to 12V charger	Orion-Tr Smart 48/12-30; fed from Lynx 48V+ bus tap through standalone F-06 (30A 58V MIDI interim, 20A 80V FKS/ATO final); Lynx Slot 4 remains open	20, 11, 133, 182
Monitoring	Cerbo GX + SmartShunt 300A	22, 23
Alternator kit	Mechman 48v secondary alternator kit with WS500	168
Load-dump clamp	Balmar APM-48	169
Alternator branch fuse	F-04 150A MEGA (58V/80V) in Lynx Slot 3	170
WS500 low-current fuse set	F-12 regulator power + F-13 positive voltage sense; current-sense pair is unfused per current manual	171
WS500 Upfitter #3 control kit	F-15 + control wire + holder/terminals	176

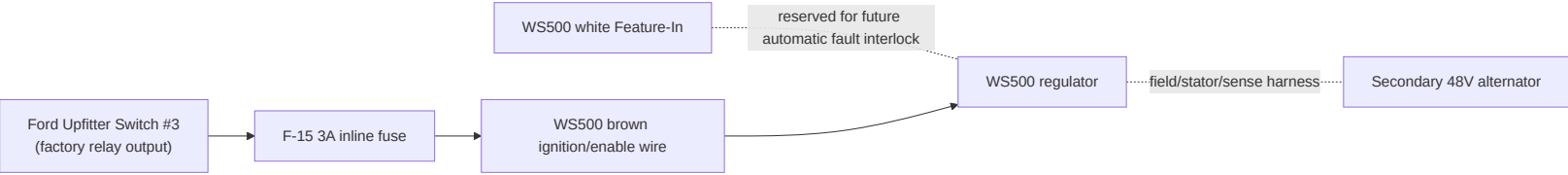
Battery manual limits and charger-programming baseline

- Battery model basis: Dumfume 51.2V 100Ah LiFePO4, 3x in parallel (1S3P ; manual allows up to 1S4P).
- Per-battery charge voltage: 58.4V +/-0.2V ; charge-limit/over-charge protection voltage also listed as 58.4V .
- Per-battery current references: recommended charge 20A ; max continuous charge 100A ; recommended discharge 50A ; max continuous discharge 200A .
- Bank-level reference for current 3x setup: recommended charge 60A ; max continuous charge 300A ; recommended discharge 150A ; max continuous discharge 600A .
- Protection thresholds: over-discharge protection 36.8V , recovery 43.2V ; discharge overcurrent protection 600A ; short-circuit protection 1800A .
- Temperature thresholds: low-temp charge protection approx 41F-50F with recovery at 50F ; high-temp protection listed as 157F / 140F .
- MultiPlus-II charger limit is 35A , so full-output charging is below the bank's 60A recommended-charge reference (~11.7A per battery in 3P). For first household-outlet tests, keep AC input current at 10A , then 12A max on a normal 15A source. Current charger-programming target is 56.8V absorption/charge, 54.0V float, 52.8V storage if shown, 0.5h max absorption, equalization off, and temperature compensation off.

48V power path



Alternator control path



Control logic

- Upfitter #3 ON : WS500 brown wire is energized, regulator is allowed to run, alternator charging can occur.
- Upfitter #3 OFF : WS500 is disabled through the brown wire, regulator field output collapses, and alternator charging stops.
- Use Upfitter #3 as the manual alternator-charge shutdown control.
- Do not use the main 48V disconnect as the first alternator shutdown method while the engine is running and the secondary alternator is charging.

Why Upfitter #3

- It is already a factory-switched, relay-driven 12V control source.
- It keeps the WS500 control in the truck cab without adding a separate aftermarket switch.
- It is suitable for a low-current enable signal, but not for any high-current alternator or battery conductor.
- A local inline fuse is still required because the factory upfitter circuit fuse is much larger than the small-gauge WS500 control wire.

Fusing and wire baseline

ID	Function	Locked value	Wire basis
F-01A/B/C	Battery branch positive protection	200A Class T provisional	2/0 AWG
F-04	Alternator branch into Lynx Slot 3	150A MEGA (58V/80V)	2/0 AWG
F-05	Lynx Slot 4	Open/blank spare fused position; not used for Orion	N/A
F-06	Orion 48v input from Lynx bus tap	30A 58v MIDI interim; 20A 80v FKS/ATO final	6 AWG planned; keep source-side unfused tap short
F-12	WS500 regulator power lead	10A baseline (15A if required by alternator case); voltage rating must cover the connected alternator/system positive voltage	harness lead
F-13	WS500 positive voltage-sense lead	3A; fuse/holder voltage class must cover the 48v bank maximum unless proven by WS500 harness documentation	harness lead
CERB0-PWR	Cerbo GX low-current power feed	1A-3A inline fuse/holder rated for the 48v bank maximum; system/load side of main disconnect preferred for bench shutdown	18 AWG red/black duplex acceptable
WS500 current-sense pair	Purple/grey current-sense high/low to shunt/current-sense point	No separate fuse position in current Wakespeed manual; twist pair if extended and route away from noise	harness lead
F-15	Upfitter #3 to WS500 brown ignition wire	3A inline ATO/ATC on 12V control circuit	16 AWG TXL/GXL control wire

Major conductors

- Battery branch and main 48V trunk wiring: 2/0 AWG .
- Parallel battery-current sharing target is similar **total loop resistance** per battery path, not equal positive-only length. The current bench layout may use short/medium/long positive leads balanced by long/medium/short negative leads; do not add unnecessary cable coils solely to make positive leads identical.

- Secondary alternator positive path (ALT B+ -> APM-48 -> F-04 -> Lynx): 2/0 AWG , ~20 ft one-way planning basis.
- Secondary alternator dedicated negative return (ALT B- -> Lynx -): 2/0 AWG , ~20 ft one-way planning basis.
- Orion 48V feeder and MPPT battery leads: 6 AWG ; Orion positive leaves the Lynx bus through standalone F-06 , not Lynx Slot 4.
- Upfitter #3 control lead to WS500 brown wire: 16 AWG TXL/GXL planning basis, ~6 ft one-way assumed until measured.

APM-48 wiring intent

- Mount the APM-48 at the alternator end of the 48V branch, as close to the alternator output as practical.
- Connect APM red to alternator B+ .
- Connect APM black to alternator B- , or to the approved alternator ground/case point if the alternator is not isolated-ground.
- Do not stack the APM ring terminals under the main battery cable lugs unless the product instructions for the exact unit explicitly allow it.

Normal operating sequence

1. Main 48V disconnect closed.
2. Upfitter #3 switched ON .
3. Engine running.
4. WS500 enabled and regulating.
5. Alternator charges the house bank through APM-48 and F-04 .

Fault and shutdown sequence

If a battery pack trips or an alternator fault is suspected:

1. Switch Upfitter #3 OFF to disable the WS500 .
2. Wait for alternator charge current to collapse.
3. Open the main 48V disconnect only after alternator charging is no longer active, if full house shutdown is required.

Reason:

- The regulator should be shut down first.
- The APM-48 is a protection layer, not the primary shutdown method.

One-battery-trip behavior

- If one battery BMS disconnects but the other two remain online, the 48V bus should stay up.
- The system does not immediately become a full-bank load dump event just because one battery disappears.
- The practical effect is that charge and discharge current redistribute to the remaining batteries.
- The real hazard is a cascade where the second and third battery also disconnect under active alternator charge.

What is still not fully closed

- Final Mechman kit fitment/content confirmation for the exact truck.
- Final WS500 harness polarity confirmation (PH vs NH).
- Official vendor confirmation that the documented Dumfume battery/BMS behavior is acceptable with the WS500 .
- Exact alternator negative/case isolation behavior in the installed Mechman kit.
- Whether future automatic fault-interlock logic should be added on the WS500 Feature-In or through an external relay.

Rule for future edits

- Update this file first for any 48V architecture, alternator-control, or shutdown-strategy change.
- Keep the implementation files for wiring detail and fuse placement, not for high-level architecture decisions.